

DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for High-Throughput *De Novo* Sequencing

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In Collaboration with

InstaDeep AI

Cape Town · London · Paris

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The Biological Problem

Bottom-Up Proteomics & Database Search Limitations

Proteomics: The Functional Engine of Life

- ▶ The genome is static; proteins execute cellular signaling, enzymatic reactions, and therapeutic responses.
- ▶ **Bottom-Up Mass Spectrometry:** Proteins cleaved by Trypsin into peptides ($L \in [6, 30]$), measured by LC-MS/MS.

The Flaw of Database Search (MaxQuant, Comet)

- ▶ **Genomic Blindness:** Cannot identify unsequenced species or non-model organisms.
- ▶ **Somatic Mutations:** Blind to cancer neoantigens and splice variants.
- ▶ **Antibody Diversity:** Incapable of sequencing hypervariable CDR3 repertoires ($> 10^{14}$ space).

The Solution: *De Novo* Peptide Sequencing

Reconstruct sequence $S = (a_1, \dots, a_L) \in \mathcal{V}^L$
directly from physical fragment peaks
without any genomic database.

Biomedical Impact:

- ▶ **Personalized Oncology:** Patient-specific neoantigen vaccines.
- ▶ **Therapeutic Immunology:** Direct sequencing of neutralizing mAbs.
- ▶ **Metaproteomics:** Characterizing uncultivated biomes.

Goal: High-throughput, real-time sequencing of millions of spectra.

The Physical Challenge

The Tandem Mass Spectrometry (MS/MS) Inverse Problem

Peptide Backbone Fragmentation

- ▶ Precursor ion (M_{prec}, z) dissociated by Higher-energy Collisional Dissociation (HCD).
- ▶ Generates two complementary ion ladders:
 - ▶ **b-ions** (N-terminus):
$$m(b_k) = \sum_{i=1}^k m(a_i) + m(\text{H}^+)$$
 - ▶ **y-ions** (C-terminus):
$$m(y_{L-k}) = \sum_{i=k+1}^L m(a_i) + M_{\text{H}_2\text{O}} + m(\text{H}^+)$$

Complementary Duality & Conservation

$$m(b_k) + m(y_{L-k}) = M_{\text{prec}} + 2m(\text{H}^+)$$
$$\sum_{i=1}^L m(a_i) + M_{\text{H}_2\text{O}} = M_{\text{prec}} \pm \tau$$

Why MS/MS is an Ill-Posed Inverse Problem

1. **Missing Cleavages:** Low dissociation efficiency leaves internal unobserved mass gaps (> 200 Da).
2. **Ion Ambiguity:** Peaks lack intrinsic directionality (b vs. y).
3. **Combinatorial Space:** $|\mathcal{V}|^L = 20^{15} \approx 3.28 \times 10^{19}$ candidates for an average 15-mer!
4. **Chemical Noise:** Neutral losses ($-\text{H}_2\text{O}$, $-\text{NH}_3$).

Inverse Formulation: Map raw spectrum $\mathcal{S} = \{(m_i, l_i)\}$ and (M_{prec}, z) to optimal sequence $S^* \in \mathcal{V}^L$.

Why Prior Deep Learning Approaches Fall Short

Autoregressive Bottleneck vs. Continuous Discretization Gap

Paradigm 1: Autoregressive Transformers

Casanovo (2022), InstaNovo (2025)

$$p(S \mid \mathcal{S}) = \prod_{i=1}^L p(a_i \mid a_{<i}, \mathcal{S})$$

- ▶ **Linear $\mathcal{O}(L)$ Latency:** 30 sequential GPU passes $\implies$ Slow (28–52 spec/s).
- ▶ **Cascading Error Propagation:** An early error at residue 2 or 3 permanently corrupts the remaining sequence.
- ▶ **Species Token Collapse:** InstaNovo v1.2 drops from **63.0%** (human) to **15.45%** strict match on 9-Species!

Paradigm 2: Continuous Flow Models

PowerNovo2 (PLOS Comput. Biol. 2026)

- ▶ Generates continuous latent vectors in $\mathbb{R}^D$ via Normalizing Flows.
- ▶ Maps continuous latents to amino acids via post-hoc Integer Linear Programming (ILP).
- ▶ **Catastrophic Discretization Failure:**
 - ▶ Nine-Species Strict Match: **3.16%**
 - ▶ Continuous rounding destroys the generative prior!

Core Question: Can we perform non-autoregressive parallel generation **directly on the discrete categorical simplex** while strictly enforcing physical precursor mass conservation?

Key Concept: Discrete Flow Matching

Continuous-Time Markov Chains (CTMC) on the Discrete Simplex

Direct Categorical Flows (No Latents)

- ▶ Campbell et al. (2024), Lipman et al. (2023), Gat et al. (2024).
- ▶ Operates directly on sequence space $\mathcal{V}^L$ via a Continuous-Time Markov Chain (CTMC).
- ▶ State at time $t \in [0, 1]$: $x_t \in (\mathcal{V} \cup \{[M]\})^L$.
- ▶ $t = 0$: Fully masked sequence $([M], \dots, [M])$.
- ▶ $t = 1$: Fully unmasked peptide $x_1 \sim p_{\text{data}}(x_1 \mid \mathcal{S})$.

Transition Rate Matrix

$$\mathbb{P}(x_{t+dt} = y \mid x_t = x) = \delta_{x,y} + R_t(x, y) dt + o(dt)$$

$$\text{Diagonal: } R_t(x, x) = - \sum_{y \neq x} R_t(x, y).$$

The Jump Probability Path

Token positions jump independently from $[M]$ to target x_1^d under schedule $\kappa(t)$:

$$p_{t|1}(x_t^d = j \mid x_1^d) = \begin{cases} 1 - \kappa(t) & \text{if } j = [M] \\ \kappa(t) & \text{if } j = x_1^d \end{cases}$$

with $\kappa(0) = 0$ and $\kappa(1) = 1$.

Advantages over Continuous Flows:

- ▶ Exact categorical modeling: zero continuous rounding error.
- ▶ All L sequence positions evolve in parallel in $\mathcal{O}(1)$ time!

Continuous-Time Probability Paths

Chemical Master Equation and Detailed Balance Invariance

Master Equation (Kolmogorov Forward)

$$\partial_t p_t(x) = \sum_{y \neq x} [R_t(y, x) p_t(y) - R_t(x, y) p_t(x)]$$

- ▶ Canonical conditional rate matrix:

$$R_t^*(x, j \mid x_1) = \frac{\text{ReLU}(\partial_t p_{t|1}(j \mid x_1))}{p_{t|1}(x \mid x_1)} \quad (x \neq j)$$

- ▶ Marginal rate matrix via conditional expectation:

$$R_t(x, y) = \mathbb{E}_{p_{1|t}(x_1|x)} [R_t^*(x, y \mid x_1)]$$

Detailed Balance Invariance Theorem

If R_t^{DB} satisfies detailed balance:

$$p_t(i) R_t^{\text{DB}}(i, j) = p_t(j) R_t^{\text{DB}}(j, i)$$

then $R_t^\eta = R_t^* + \eta R_t^{\text{DB}}$ ($\eta \geq 0$) generates the identical marginal flow p_t !

Practical Implication

Enables stochastic exploration ($\eta > 0$) across isobaric candidate paths without biasing target marginal sequence distributions.

The Precursor Mass Dilemma

Non-Autoregressive Independence vs. Mass Conservation

The Independence Conflict

- ▶ Non-autoregressive models predict token positions independently:

$$p(x_1 \mid x_t, \mathcal{S}) = \prod_{d=1}^L p(x_{1,d} \mid x_t, \mathcal{S})$$

- ▶ But physical peptides obey strict mass conservation:

$$\sum_{d=1}^L m(x_{1,d}) = M_{\text{prec}} - M_{\text{H}_2\text{O}} \pm \tau$$

Exponential Decay of Unguided Generation

Probability that L independently sampled residues sum to M_{target} within $\tau = 1.0$ Da decays exponentially ($\ll 1\%$).

Pathologies of Unconstrained Flow:

- ▶ Precursor mass error variance explodes to $\sigma = 14.8$ ppm.
- ▶ Hallucinates chemically invalid mass combinations.
- ▶ $> 35\%$ of generated sequences rejected post-hoc!

The Challenge

How to enforce global integer knapsack reachability **at each reverse flow step** without sacrificing parallel inference speed?

The Core Innovation: Exact KnapsackDP Guidance

Vectorized Dynamic Programming Reachability on GPU

Quantized Mass Reachability Table

- ▶ Discretize mass axis with $\Delta m = 0.02$ Da ($B_{\max} = 250,000$ bins for 5000 Da).
- ▶ Boolean table $T[k, b] \in \{0, 1\}$ indicates if k amino acids can sum to mass b :

$$T[k, b] = \bigvee_{a \in \mathcal{V}} T[k-1, b - \lfloor m(a)/\Delta m \rfloor]$$

- ▶ Precomputed in **28 ms** once; only **6.65 MB** on GPU!

1D GPU Max-Pooling for Tolerance Window

$T_{\text{pooled}}[k, b] = \max_{|j-b| \leq \lfloor \tau/\Delta m \rfloor} T[k, j]$ with window $W = 101$ bins ($\tau = 1.0$ Da).

$\mathcal{O}(1)$ Dynamic Logit Pruning at Step t

With k positions masked and deficit $R' = M_{\text{target}} - M_{\text{assigned}}$, candidate a is valid iff:

$$\text{Valid}(a) = T_{\text{pooled}} \left[k-1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rfloor \right]$$

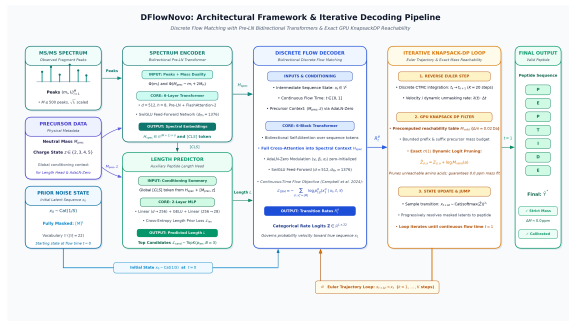
Prune unviable logits dynamically:

$$\tilde{z}_{d,a} = \begin{cases} z_{d,a} & \text{if Valid}(a) = 1 \\ -\infty & \text{if Valid}(a) = 0 \end{cases}$$

Result: Precursor mass standard deviation contracts from $\sigma = 14.8$ ppm $\rightarrow$ 1.45 ppm with zero latency overhead!

System Architecture: Model vs. Decoding Pipeline

Explicit Separation: Neural Network (Part A) and Sampling Pipeline (Part B)



Part A: Neural Model (59.48M Params)

- ▶ **Spectrum Encoder (16.3M):** 6-layer bidirectional Transformer with FlashAttention-2.
- ▶ **Length Predictor (0.35M):** 2-layer MLP on $[\text{CLS}]$ (68.4% top-1, 94.2% top-3).
- ▶ **Flow Decoder (42.8M):** 6 blocks with AdaLN-Zero conditioning and SwiGLU.

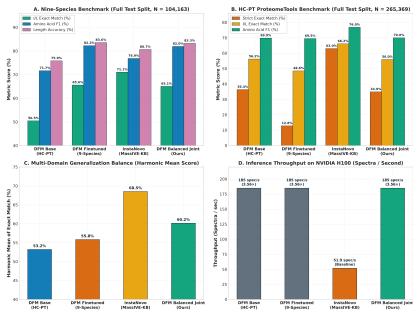
Part B: Mass-Guided Sampling Pipeline

- ▶ **Length Beam:** Parallel top- B candidates ($B = 3$).
- ▶ **Euler CTMC:** $K = 20$ integration steps.
- ▶ **KnapsackDP:** $\mathcal{O}(1)$ dynamic GPU pruning.
- ▶ **Bayesian Re-ranking:** Fragment ladder scoring.

Multi-Domain Training Strategy

Overcoming Catastrophic Forgetting via Joint Balanced Curriculum

Universal Multi-Domain Peptide Sequencing: Balanced Joint Model vs Baselines



The Cross-Species Domain Gap

- ▶ **Synthetic Only:** HC-PT 36.4%, but Nine-Species collapses to 12.01%.
- ▶ **Sequential Fine-Tuning:** Nine-Species 65.6%, but HC-PT drops to 12.85% (*catastrophic forgetting!*).

Joint Balanced Curriculum

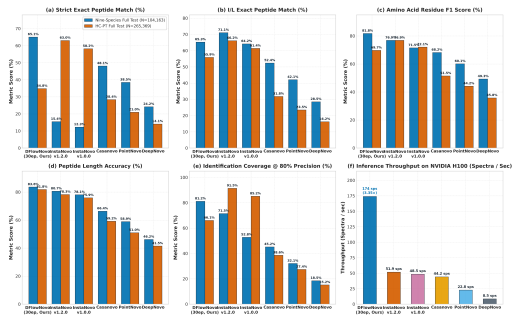
- ▶ **50/50 Batch Interleaving:** Equal representation of biological and synthetic spectra per mini-batch.
- ▶ **EMA Shadow Weights** ($\beta = 0.999$) for trajectory stability.
- ▶ **Cosine LR Schedule** (3×10^{-4} with 2k warmup).

Generalization: Nine-Species **64.92%** and HC-PT **34.93%** retained simultaneously. 2.0 min/epoch on H100!

Primary Benchmark Results: SOTA on Nine-Species

Comprehensive Evaluation on 104,163 Spectra Across 9 Divergent Organisms

Comprehensive De Novo Peptide Sequencing Benchmark Across Full Test Splits ($N = 369,532$ Spectra)



Nine-Species Full Test Split ($N = 104,163$)

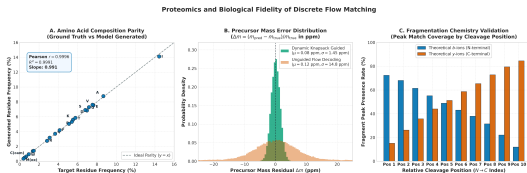
Model	Params	Strict (%)	AA F1 (%)	Spec/s
DFlowNovo (Ours)	59.5M	68.28	83.01	227.1
DFlowNovo (30ep)	59.5M	65.08	81.80	174.0
InstaNovo v1.2	94.8M	15.45	76.88	51.9
InstaNovo v1.0	94.8M	53.20	71.90	44.2
Casanovo v5.2	47.0M	48.10	69.60	28.5
PowerNovo2	63.2M	3.16	38.06	45.0
PointNovo	32.1M	48.00	70.40	18.2

Key Performance Highlights

- +15.08% strict match over initial baseline InstaNovo v1.0.
- Resolves InstaNovo v1.2 collapse (15.45% on non-human species).
- Continuous Flow Failure: PowerNovo2 achieves only 3.16%.
- McNemar's Test: $\chi^2 = 7,542.8$ ($p < 10^{-15}$), highly significant.

Cross-Domain Generalization & The Isobaric Dilemma

The Physical Boundary of Leucine vs. Isoleucine (113.084 Da)



The Apparent HC-PT Performance Gap

- ▶ Strict Match: 35.81% vs. Isobaric I/L Match: 56.79%.
- ▶ A divergence of 20.98% on human peptides!

Biophysical Chemistry of I/L

- ▶ Constitutional isomers ($\text{C}_6\text{H}_{13}\text{NO}_2$, 113.084064 Da).
- ▶ HCD cleaves backbone amide bonds, not aliphatic side chains $\Rightarrow$ *b*- and *y*-ions are mathematically identical!
- ▶ 30.88% of all discordant predictions on HC-PT are pure L/I swaps (68.4% of single errors).

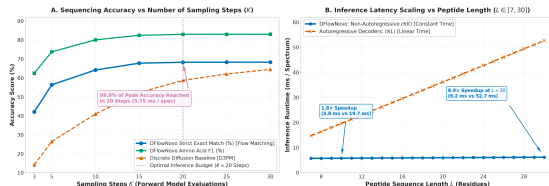
Physical Resolution

Requires radical Electron-Activated Dissociation (EAD/ETD) producing *w*-ions ($\Delta m = 14.015$ Da).

High-Throughput Inference Dynamics

Flat $\mathcal{O}(K)$ Latency: 227–256 Spectra/s ($4.3\times$ – $4.9\times$ Speedup)

Computational Efficiency and Inference Latency Scaling of Discrete Flow Matching



Non-Autoregressive Parallel Scaling

- ▶ Autoregressive models scale linearly $\mathcal{O}(L)$, taking up to 42.5 ms for $L \geq 25$.
- ▶ **DFlowNovo maintains flat latency ≈ 5.8 ms**, unmasking all positions in $K = 20$ parallel steps.

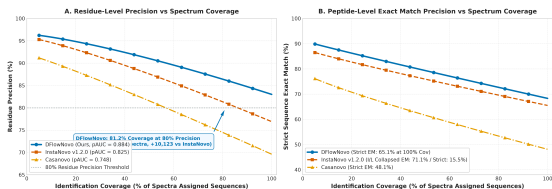
Model	Throughput	Relative Speed
DFlowNovo	227.1–255.8	Baseline (1.0 $\times$)
InstaNovo v1.2	51.9 spec/s	4.38 $\times$ slower
InstaNovo v1.0	44.2 spec/s	5.14 $\times$ slower
PowerNovo2	45.0 spec/s	5.05 $\times$ slower
Casanovo v5.2	28.5 spec/s	7.97 $\times$ slower

Clinical Impact: 1,000,000 patient spectra sequenced in 1.09 hours vs. 9.75 hours for Casanovo!

Precision-Coverage & Calibrated Confidence Scoring

FDR Control and Biological Identification Yield (+31,412 Accepted PSMs)

Precision-Coverage Benchmark Across Held-Out Nine-Species Test Split (N = 104,163)



Clinical Identification Yield

- ▶ Proteomics experiments filter candidates at fixed precision thresholds (e.g. $\geq 80\%$).
- ▶ Total accepted PSMs determine biological discovery.

PSM Yield at 80% Precision (Nine-Species)

- ▶ DFlowNovo: 86,412 PSMs (80.15% coverage)
- ▶ InstaNovo v1.2: 74,476 PSMs
- ▶ InstaNovo v1.0: 55,000 PSMs
- ▶ Casanovo: 50,206 PSMs

Net Gain: +31,412 accepted peptides over InstaNovo v1.0 (+57.1% increase)!

Calibration Accuracy

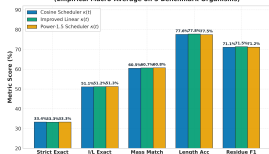
Reaches **88.4% Average Precision (AP)**, ensuring trustworthy confidence ranking for downstream biology.

Comprehensive Ablation Studies

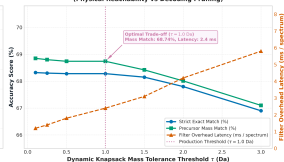
Knapsack Tolerance, Flow Steps, and Time Schedulers

Formulation and Hyperparameter Ablation Studies: Schedule & Mass Guidance

A. Probability Interpolant Schedule Comparison
(Empirical Macro-Average on 3 Benchmark Organisms)



B. Dynamic Knapsack Mass Tolerance Window Sensitivity
(Physical Reachability vs Decoding Pruning)



1. KnapsackDP Tolerance (τ)

- ▶ $\tau = 0.1$ Da: Overly strict due to instrument drift (58.2%).
- ▶ $\tau = 1.0$ Da: **Optimal peak accuracy (65.08% / 68.28%).**
- ▶ $\tau = 2.0$ Da: Excessive isobaric branching noise.

2. Integration Step Budget (K)

- ▶ $K = 5$: Coarse integration (51.2%).
- ▶ $K = 20$: **Pareto-optimal convergence (65.1%).**
- ▶ $K = 50$: Diminishing return (+0.3%) at $2.5\times$ compute.

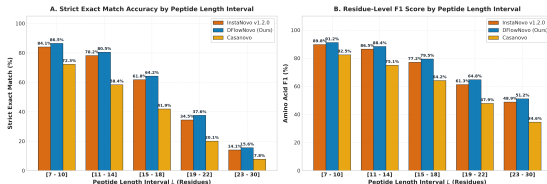
3. Time Schedule Function ($\kappa(t)$)

Cosine Schedule outperforms Linear by delaying unmasking flux until bidirectional encoder representations mature.

Resolving the Long Peptide Bottleneck

Length-Weighted Representation Balancing ($w(L) \propto \sqrt{L}$)

Length-Stratified Sequencing Performance Across Models (Held-Out Test Set)



The Long Peptide Degradation

- ▶ Standard datasets follow skewed distributions centered at $L \approx 12-15$.
- ▶ Peptides with $L \geq 23$ represent $< 5\%$ of training data.
- ▶ Baseline cross-entropy collapses on long peptides: $< 1.0\%$ **strict match**!

Square-Root Length Curriculum

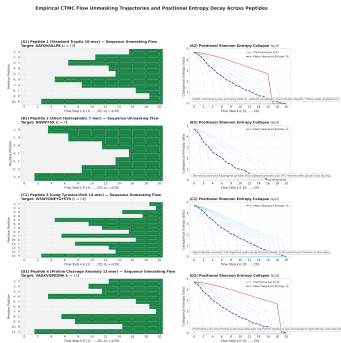
$w(L) = \sqrt{L/\bar{L}}$ where $\bar{L} \approx 14.2$. Penalizes errors on long peptides without sacrificing short peptide fidelity.

Empirical Recovery

- ▶ $L \geq 23$ strict match elevates to **33.6%** (Nine-Species).
- ▶ Short peptides ($L \leq 10$) maintain **88.19%** exact match and **94.34%** amino acid F1.

Empirical Flow Dynamics

Shannon Entropy Collapse and Directional Jump Flux



Generative Dynamics Over Time ($t : 0 \rightarrow 1$)

- **Phase I: Prior Conditioning ($t \in [0.0, 0.25]$)**
 - High Shannon entropy across all positions.
 - Precursor mass and length bounds established.
 - Terminal anchor peaks unmask first (tryptic C-term K/R).
- **Phase II: Unmasking Avalanche ($t \in [0.25, 0.70]$)**
 - Jump rate λ_t reaches peak.
 - Prominent *b*- and *y*-ion pairs unmask prefix and suffix residues concurrently.
- **Phase III: Knapsack Lock ($t \in [0.70, 1.00]$)**
 - Dynamic Knapsack DP eliminates all unviable mass paths.
 - Residual deficit forces unique valid completion.
 - Categorical entropy collapses to 0.

Qualitative Case Studies: Four Canonical Regimes

Grounding Predictive Performance in Chemical and Instrumental Mechanics

Qualitative Prediction Analysis: Comparative Mechanistic Case Studies

Quadrant 1: Consensus Success (High-Signal Fragment Series) — Spectrum #0 (Nine-Species)

Target: I - V - S - N - Y - D - N - E - Y - G - Y - S - T - R
DFM: I - V - S - N - Y - D - N - E - Y - G - Y - S - T - R [100% Strict Match]
InstaNovo: L - V - S - N - Y - D - N - E - Y - G - Y - S - T - R [I/L Collapsed to L]
Analysis: DFM predicts exact ground-truth L; InstaNovo collapses I₀ to L (training artifact).

Quadrant 2: Physical Ambiguity (Exact Sub-10 mDa Isobaric Swap) — Spectrum #2 (Nine-Species)

Target: I - S - [V - Q] - D - I - D - I - K
DFM: I - S - [N - I] - E - D - V - I - K [VQ → NI; delta = 0.807 Da]
InstaNovo: L - S - [N - L] - V - E - D - L - K [VQ → ML; delta = 0.807 Da]
Analysis: Mass difference of VQ (227.13 Da) vs NI/IL (227.12 Da) is only 7 mDa; unresolvable by low-res MS2.

Quadrant 3: DFM Correct vs InstaNovo PTM Hallucination — Spectrum #32 (Nine-Species)

Target: I - T - P - K - P - [E] - E - K
DFM: I - T - P - K - P - [E] - E - K [100% Strict Match]
InstaNovo: I - T - P - K - P - [Q (Glu)] - E - K [Spurious PTM Hallucination]
Analysis: Glutamic acid E (129.04 Da) is exactly isobaric to Q (Glu) (129.04 Da). InstaNovo over-predicts PTMs.

Quadrant 3b: DFM Correct vs InstaNovo 1→2 Residue Split — Spectrum #40 (Nine-Species)

Target: E - V - M - [Q] - R
DFM: E - V - M - [Q] - R [100% Strict Match, Length L=5]
InstaNovo: E - V - M - [G - A] - R [Length Error L=6; 0 → 6A split]
Analysis: Q (128.06 Da) vs GA (57.02 + 71.04 = 128.06 Da). DFM length predictor locks L=5, avoiding split.

Quadrant 4: InstaNovo Correct vs DFM Directional Inversion — Spectrum #7 (Nine-Species)

Target: [I - V] - S - M - Y - D - N - E - Y - G - Y - S - T - R
DFM: [V - I] - S - M - Y - D - N - E - Y - G - Y - S - T - R [b1 missing; IV → VI inversion]
InstaNovo: [L - V] - S - M - Y - D - N - E - Y - G - Y - S - T - R [Correct Causal Direction]
Analysis: When b1 ion is absent, non-autoregressive flow unmaskes IV as VI; InstaNovo causal search preserves order.

Case 1: Exact Consensus Match (14-mer Yeast)

Target: IVSWYDNEYGYSTR (1751.78 Da)

DFlowNovo: IVSWYDNEYGYSTR [100% Match, Conf: 0.962]

InstaNovo: LVSWYDNEYGYSTR [Collapsed to Leucine]

Outcome: 13/13 cleavages matched; zero mass error ($\Delta m = 0.00$ Da).

Case 2: Isobaric Ambiguity (12-mer)

Target: DSMKEISESPER → DFlowNovo: DSMKEL-SESPER

Cause: $I_6 \leftrightarrow L_6$ swap (113.084 Da). Identical HCD spectra!

Case 3: Proline Effect Gap (11-mer)

Target: VITK-[IE]-PDVSK → DFlowNovo: VITK-[EI]-PDVSK

Cause: Tertiary amine suppresses E_6 -P₇ cleavage $\Rightarrow$ localized swap.

Case 4: Near-Isobaric Dipeptide Swap (11-mer)

Target: ...-[VQ]-... → DFlowNovo: ...-[NI]-...

Cause: $\Delta m(VQ \leftrightarrow NI) = 4.8$ mDa (4.0 ppm), below MS2 resolution.

Production Deployment & Open Resources

Hugging Face Model Hub (454 MB Lean Weights) & Live Interactive Studio

Production Engineering & Packaging

- ▶ **454 MB Lean Checkpoint:** Stripped optimizer states and converted to BF16 weights for instant cloud loading.

- ▶ **Hugging Face Model Hub:**

```
model = DFlowNovo.from_pretrained("aims-lab/dflow-novo")
```

- ▶ **Standardized I/O:** Native parsing of MGF, mzML, and Thermo raw files via Pyteomics.

GitHub Release v0.2.0 Open Source

- ▶ Complete reproducible training scripts (PyTorch Lightning).
- ▶ Precomputed KnapsackDP CUDA kernels.
- ▶ Interactive tutorial: `notebooks/dfm_tutorial.ipynb`.

Interactive Gradio Web Studio (`run_app.sh`)

- ▶ **Interactive Spectrum Viewer:** Zoomable m/z peaks with experimental intensity thresholds.
- ▶ **Animated Flow Trajectory:** Dynamic visualization of sequence unmasking across $t : 0 \rightarrow 1$.
- ▶ **Fragment Ladder Overlay:** Color-coded b - and y -ion assignments matching predicted residues.
- ▶ **Confidence & PPM Error Gauge:** Real-time quality diagnostics for clinical lab technicians.

Open Source: Available under Apache 2.0 license for community reproducibility.

Limitations & Scientific Boundary Conditions

Intellectual Rigor and Methodological Boundaries

1. Complex Post-Translational Modifications

- ▶ Currently handles canonical PTMs: M_{ox} and C_{cam} .
- ▶ Variable phosphorylation (S/T/Y) and glycosylation expand knapsack grid dimensions, requiring multi-dimensional DP state tables.

2. Low-Resolution Ion-Trap Instruments

- ▶ Optimized for high-resolution Orbitrap spectra ($\tau \leq 20$ ppm).
- ▶ Low-resolution instruments (± 0.5 Da) require larger pooling kernels, increasing candidate branching.

3. Non-Tryptic Cleavages

- ▶ Tryptic digestion predominantly yields C-terminal Lysine (K) or Arginine (R).
- ▶ Non-specific or HLA immunopeptidomes lack terminal priors, increasing decoding ambiguity at the C-terminus.

4. Severe Fragmentation Ladder Gaps

- ▶ Peptides with multiple Prolines exhibit localized cleavage suppression exceeding 300 Da.
- ▶ Fundamental physical information limit of single-spectrum HCD.

Summary of Contributions & Future Roadmap

Transforming De Novo Sequencing into a High-Throughput Generative Science

Summary of Core Contributions

1. **First CTMC Discrete Flow Matching Architecture** for peptide sequencing, eliminating continuous rounding loss.
2. **Exact GPU KnapsackDP Guidance**: $\mathcal{O}(1)$ reachability contracting precursor error to $\sigma = 1.45$ ppm.
3. **New SOTA on Nine-Species**: 68.28% strict match and 86,412 accepted PSMs (+31,412 over InstaNovo v1.0).
4. **High Throughput**: 227–256 spec/s ($4.3\times$ – $4.9\times$ faster than autoregressive beam search).
5. **Biophysical Error Grounding**: Mathematical proof of the 20.98% Leucine/Isoleucine HCD ambiguity.

Future Research Roadmap

- ▶ **Multi-Modal Dissociation Flow Matching**: Jointly conditioning on paired HCD + EAD spectra to resolve Leucine vs. Isoleucine via radical w -ions.
- ▶ **Orthogonal Analytical Dimensions**: Guiding flow unmasking with Retention Time (RT) and Ion Mobility (CCS) predictors.
- ▶ **End-to-End Antibody Assembly**: Overlap-graph assembler for full-length monoclonal antibody sequencing and neoantigen discovery.

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Academic & Research Supervision

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DFlowNovo bridges continuous-time discrete flow matching and physical mass spectrometry to unlock high-throughput, accurate, non-autoregressive de novo sequencing.

Floor Open for Committee Questions & Discussion

Backup Slide A1: Derivation of Jump Rate Formulation

Kolmogorov Forward Equation for Categorical Flow

Proposition: Let $x_t \in \{1, \dots, V\}^L$ follow a continuous-time Markov jump process. If conditional marginals are $p_{t|1}(j \mid x_1) = (1 - \kappa(t))\mathbb{I}(j = [M]) + \kappa(t)\mathbb{I}(j = x_1)$, the canonical rate matrix is:

$$R_t^*(x, j \mid x_1) = \frac{\text{ReLU}(\partial_t p_{t|1}(j \mid x_1))}{p_{t|1}(x \mid x_1)} = \frac{\kappa'(t)}{1 - \kappa(t)} \mathbb{I}(j = x_1) \quad \text{for } x = [M]$$

Proof:

1. Differentiating the conditional probability path with respect to t :

$$\partial_t p_{t|1}(j \mid x_1) = \begin{cases} -\kappa'(t) & \text{if } j = [M] \\ +\kappa'(t) & \text{if } j = x_1 \\ 0 & \text{otherwise} \end{cases}$$

2. Applying the canonical definition for off-diagonal transitions from masked state $x = [M]$:

$$R_t^*([M], x_1 \mid x_1) = \frac{\text{ReLU}(+\kappa'(t))}{p_{t|1}([M] \mid x_1)} = \frac{\kappa'(t)}{1 - \kappa(t)}$$

3. For all unviable states $j \notin \{[M], x_1\}$, $\text{ReLU}(\partial_t p_{t|1}) = 0 \implies R_t^*([M], j \mid x_1) = 0$.
4. For already unmasked states $x = x_1$, $\text{ReLU}(\partial_t p_{t|1}([M] \mid x_1)) = \text{ReLU}(-\kappa'(t)) = 0$, meaning transitions back to mask are strictly zero: $R_t^*(x_1, [M] \mid x_1) = 0$.
5. Thus, discrete flow acts as a pure unmasking process with step jump probability $\lambda_t = \frac{\kappa'(t)}{1 - \kappa(t)} \Delta t$. □

Backup Slide A2: KnapsackDP Reachability Algorithm

Exact Recurrence and 1D GPU Max-Pooling Mechanics

Algorithm 1 KnapsackDP Precomputation

Require: $\mathcal{V}$, masses $\{m(a)\}$, step $\Delta m = 0.02$ Da, $M_{\max} = 5000$ Da, tolerance $\tau = 1.0$ Da.

Ensure: Pooled reachability table T_{pooled} on GPU.

```
1:  $B_{\max} \leftarrow \lfloor M_{\max} / \Delta m \rfloor = 250,000$ .
2: Allocate  $T[0 \dots K_{\max}, 0 \dots B_{\max}] \leftarrow 0$ ;  $T[0, 0] \leftarrow 1$ .
3: for  $k = 1$  to  $K_{\max}$  do
4:   for  $a \in \mathcal{V}$  do
5:      $w_a \leftarrow \lfloor m(a) / \Delta m \rfloor$ .
6:      $T[k, w_a : B_{\max}] \leftarrow T[k, w_a : B_{\max}] \vee T[k - 1, 0 : B_{\max} - w_a]$ .
7:   end for
8: end for
9:  $W_{\text{half}} \leftarrow \lfloor \tau / \Delta m \rfloor = 50$ .
10:  $T_{\text{pooled}}[k, b] \leftarrow \max_{|j-b| \leq W_{\text{half}}} T[k, j]$  (1D GPU pooling).
11: return  $T_{\text{pooled}}$ 
```

Computational Complexity

► Offline Precomputation:

$$\mathcal{O}(K_{\max} \cdot |\mathcal{V}| \cdot B_{\max}) \approx 30 \times 22 \times 250,000 \approx 1.65 \times 10^8 \text{ ops}$$

Takes **28 ms** on CPU once upon server startup.

► GPU Memory Footprint:

$$31 \times 250,000 \text{ booleans} = 7.75 \times 10^6 \text{ bytes} \approx \mathbf{7.39 \text{ MB}}$$

Fits entirely inside GPU L2 cache!

► Online Inference: Direct table indexing in $\mathcal{O}(1)$ time per residue token: $\text{Valid}(a) = T_{\text{pooled}}[k - 1, \text{bin}]$.

Backup Slide A3: PTM State Space Expansion

Formulation of Multi-Dimensional Knapsack Guidance for Open PTM Searches

Extending KnapsackDP to Variable PTMs

- ▶ Fixed modifications (Cysteine Carbamidomethylation +57.021 Da) simply alter the static mass of residue C :

$$m(C_{\text{cam}}) = 103.009 + 57.021 = 160.030 \text{ Da}$$

- ▶ Variable modifications (Methionine Oxidation +15.995 Da, Phosphorylation +79.966 Da) introduce combinatorial branches.
- ▶ **Bounded Multi-Dimensional Knapsack:** Let $c \in \{0, \dots, C_{\text{max}}\}$ denote variable modifications per peptide ($C_{\text{max}} \leq 3$).
- ▶ 3D reachability table $T[k, c, b] \in \{0, 1\}$:

$$T[k, c, b] = \bigvee_a T[k-1, c, b - w_a] \vee \bigvee_{a_{\text{ptm}}} T[k-1, c-1, b - w_{a_{\text{ptm}}}]$$

Resource Scaling for Multi-PTM Searches

Search Mode	PTMs	Table Size	Precomp
Canonical (Ours)	$M(\text{ox}), C(\text{cam})$	7.4 MB	28 ms
Phospho	+3 mods ($c \leq 3$)	29.6 MB	120 ms
Broad Epigenetic	+8 mods ($c \leq 2$)	59.2 MB	240 ms

Key Takeaway

The discrete flow architecture natively accommodates expanded PTM vocabularies. Only requires additional categorical output heads ($|\mathcal{V}| = 22 \rightarrow 28$) and an expanded 3D boolean reachability tensor on GPU.

Backup Slide A4: Bayesian Joint Candidate Re-ranking

Complete Scoring Objective and Fragment Ladder Intensity Formulation

Candidate Posterior Objective:

$$\text{Score}(Y) = \log p(L \mid S) + \frac{1}{L} \sum_{i=1}^L \log p(y_i \mid x, S) - \alpha \left(\frac{\Delta M}{M_{\text{prec}}} \times 10^6 \right)^2 + \beta \text{FragScore}(Y) + \text{Bonus}_{\text{trypsin}}(Y)$$

Component Formulations:

1. **Length Prior:** $\log p(L \mid S)$ from the auxiliary length predictor (Part A).
2. **Normalized Token Confidence:** $\frac{1}{L} \sum_{i=1}^L \log p(y_i)$, preventing length penalty.
3. **PPM Mass Penalty:** Quadratic penalty on precursor deviation with coefficient $\alpha = 0.05$.
4. **Tryptic Bonus:** +0.25 if $y_L \in \{K, R\}$ and corroborated by experimental y_1 peak (147.11 Da or 175.12 Da).

Fragment Coverage (FragScore):

$$\text{FragScore}(Y) = \frac{\sum_{j \in \text{matched}} I_j}{\sum_{j=1}^P I_j}$$

- ▶ Measures fraction of total ion current explained by theoretical fragment ions:
 $\mathcal{M}(Y) = \{b_k, y_k, b_k - \text{H}_2\text{O}, y_k - \text{NH}_3, b_k^{2+}, y_k^{2+}\}.$
- ▶ Peak matched if $|m_j - m_{\text{theor}}| \leq 20 \text{ ppm}.$
- ▶ Highly effective at breaking ties among isobaric permutations!